

E-tiGo Fleet Management — Complete Reference

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Status: Phase 1.5 Complete — API ready for integration (611 tests passing, 37 fleet-specific)

Context: Following the Chairman's directive to align with industry standard practices, this document consolidates all fleet management decisions, industry justifications, technical implementation, and developer handoff into a single reference.

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Part I — Decisions & Industry Standard Practices

1. Executive Summary

E-tiGo's fleet management module enables the company to own vehicles and assign them to drivers under a hire-to-own (remittance) model. This is the dominant fleet model in Nigeria's ride-hailing and taxi industries — the same approach used by Bolt fleet partners, Uber fleet operators, InDrive fleet managers, and traditional taxi operators nationwide.

Every technical and business decision documented below follows industry-proven patterns. Where the client's original direction diverged from standard practice, we have adjusted per the Chairman's instruction to follow

industry norms while clearly explaining why.

Decision Summary Table

#	Decision	Standard Practice Alignment
1	Remittance-first per-ride model	Standard Nigerian fleet/taxi model (Bolt, Uber, InDrive, traditional)
2	Zero commission for fleet drivers	Standard fleet structure (commission is embedded in remittance)
3	Tiered 3-7-14 day shortfall escalation	Standard graduated response (fleet, finance, HR)
4	Full price transparency for drivers	Required by Hire Purchase Act and FCCPA
5	Financial completion ≠ ownership transfer	Standard hire-purchase practice; legally required separation
6	Vehicle swap as terminate + recreate	Standard fleet operation with audit trail preservation
7	Settlement tracking on termination	Required by hire-purchase law; standard in vehicle finance
8	4:00 AM daily settlement	Standard fintech/ride-hailing settlement window
9	Full audit trail	Regulatory requirement (CBN, FCCPA)
10	Build when ready, not "code and deactivate"	Standard software engineering practice

2. Remittance-First Per-Ride Model

Decision

Each ride fare is allocated in real-time: the first portion goes toward the driver's daily remittance target (e.g., ₦40,000). Once the target is met, all subsequent earnings go directly to the driver. No commission is charged to fleet drivers.

How It Works

Daily Target: ₦40,000
Ride 1: ₦3,200 fare → ALL to E-tiGo (₦3,200/40,000)
Ride 2: ₦5,800 fare → ALL to E-tiGo (₦9,000/40,000)
...
Ride 9: ₦6,000 fare → ALL to E-tiGo (₦38,000/40,000)
Ride 10: ₦5,500 fare →
₦2,000 to E-tiGo (target met)
₦3,500 to DRIVER
After target met:
Ride 11: ₦4,200 → ALL to driver

Ride 12: ₦3,800 → ALL to driver

Key concept: The driver earns ₦0 until the daily target is met. After that, every fare is 100% theirs.

Industry Standard

This is the **standard Nigerian fleet/taxi remittance model**. It is used by:

- **Bolt fleet partners** — fleet owners set a daily/weekly target; drivers remit first, keep the rest
- **Traditional taxi operators** — daily "owo oga" (owner's fee) paid before driver earns
- **Uber fleet partners** — similar remittance structure with daily targets
- **InDrive fleet managers** — daily target-based deductions

The model is universally understood by Nigerian commercial drivers and aligns with their expectations when joining a fleet.

Why This Over End-of-Day Deduction

Aspect	End-of-Day (proposed)	Remittance-First (decided)
When deduction happens	Once, at day's end	Accumulated per ride throughout the day
Driver sees earnings during day	Full gross, deduction happens later	Running tracker: "₦28,000 of ₦40,000 remitted"
What driver "keeps"	Gross minus deduction at night	Only earnings AFTER target is met
Driver psychology	"I earned ₦25K but ₦3.5K was taken"	"I need to hit ₦40K, then I earn for myself"

End-of-day batch deduction creates a different psychology: drivers feel "money was taken from them." The remittance-first model is transparent — the driver knows exactly where they stand throughout the day and experiences the moment they "start earning for themselves" as a motivating milestone. This is why every major Nigerian fleet operator uses remittance-first.

Implementation Status: Done

- Per-ride fare allocation: remittance portion vs. driver earnings
- Real-time progress tracking (driver sees "₦28,000 of ₦40,000 remitted")
- Daily target configurable per agreement (different vehicles can have different targets)
- Automatic agreement completion when total vehicle cost is fully remitted
- Fleet remittance wired into ride completion flow (`ProcessPaymentJob`)

3. Zero Commission for Fleet Drivers

Decision

Fleet drivers pay 0% platform commission. E-tiGo's revenue comes entirely from the daily remittance.

Industry Standard

This is the **standard practice for fleet-owned vehicles in ride-hailing**:

- **Bolt Fleet** — fleet partners (vehicle owners) pay Bolt a platform commission; their drivers pay the fleet partner a remittance. The driver does not also pay the platform a commission.
- **Uber Fleet** — same structure. The platform's cut comes from the fleet partner, not the individual fleet driver.
- **Traditional taxi** — the driver pays the owner's daily fee (remittance). There is no additional "commission" on top.

Charging both a remittance AND a commission would be non-standard and would make fleet driving economically unviable. The daily remittance target already incorporates vehicle cost, insurance, maintenance, and E-tiGo's margin.

Implementation Status: Done

- Commission rate automatically returns 0% for any driver with an active fleet agreement
 - Enforced at the service layer (`CommissionService`) — cannot be bypassed by configuration
-

4. Tiered Shortfall Escalation (3-7-14 Days)

Decision

When a driver fails to meet their daily remittance target, the system tracks consecutive shortfall days and escalates through three tiers:

- **3 consecutive days** → Warning flag
- **7 consecutive days** → Review flag (admin attention required)
- **14 consecutive days** → Escalated flag (action required)

Industry Standard

Tiered escalation is **standard practice in fleet and vehicle finance**:

- **Bolt fleet partners** — fleet owners typically have their own 3-5 day warning followed by driver reassignment or vehicle recall
- **Vehicle hire-purchase (Nigeria)** — missed payments trigger escalating notices before repossession (typically 7-14 day windows per the Hire Purchase Act)
- **Microfinance vehicle loans** — tiered default tracking with graduated response

A single fixed threshold (e.g., "action after 7 days") does not account for the difference between a temporary slow period and a persistent pattern. The tiered approach gives E-tiGo graduated data to make proportionate decisions.

Excused Days

Not all shortfall days are the driver's fault. Industry standard practice is to account for:

- **Approved leave** — driver requested and received approval for time off
- **Vehicle downtime** — vehicle in maintenance or repair
- **Low demand** — genuinely low ride demand (admin discretion)
- **Payment system failures** — technical issues preventing ride completions

Excused days do not count toward the shortfall streak. This prevents unfair escalation and is standard in any performance-based fleet or employment system.

Implementation Status: Done

- Automatic shortfall streak tracking per agreement

- Configurable escalation thresholds (3/7/14 days, adjustable via `config/fleet.php`)
 - Excused-day support with reason categories
 - Admin endpoint to mark a day as excused (decrements streak by 1)
 - `RemittanceShortfallAlertJob` runs daily at 4:00 AM WAT to flag agreements at each tier
 - Audit log for all escalation changes
-

5. Price Transparency for Drivers

Decision

Drivers now see the full financial terms of their agreement: total vehicle cost, total remitted, remaining amount, and progress percentage. The UI should emphasise percentage progress but the financial details must be accessible.

Industry Standard

Full financial disclosure is a legal requirement for hire-purchase agreements in Nigeria:

- **Hire Purchase Act (Laws of the Federation)** — requires disclosure of the cash price, total hire-purchase price, and all payment terms
- **Federal Competition and Consumer Protection Act (FCCPA) 2018, Sections 123-127** — requires disclosure of material terms in credit and instalment purchase agreements
- **CBN Consumer Protection Framework** — requires price disclosure in consumer financial products

Every reputable hire-purchase provider in Nigeria — from vehicle finance companies to mobile phone instalment plans (e.g., MTN Device Financing, 9Mobile handset plans) — discloses the total cost to the buyer. Hiding the total price would:

1. Violate the Hire Purchase Act
2. Create legal exposure if any driver disputes the arrangement
3. Undermine trust with drivers

What Changed

The original decision (Q7) was to hide the vehicle price and show only percentage progress. This was **reversed** per the Chairman's directive to follow standard practice. The engineering team had previously flagged this as a legal risk — the reversal aligns E-tiGo with industry norms.

Implementation Status: Done

- All financial terms (`total_vehicle_cost` , `remaining_amount` , `total_remitted` , `progress_percentage`) visible to drivers via API
 - Admin sees additional operational fields (shortfall flags, settlement details) that are not relevant to drivers
-

6. Agreement Completion vs. Ownership Transfer

Decision

When a driver's total remittance reaches the vehicle's total cost, the agreement status changes to "Financially Complete." This does **not** automatically transfer legal ownership of the vehicle.

Industry Standard

Separating financial completion from ownership transfer is standard in hire-purchase:

- **Hire Purchase Act** — ownership transfers only when "the property in the goods shall pass to the hirer" upon completion of ALL terms, not just payment
- **Vehicle finance companies** — final payment triggers a process: title transfer, re-registration with DVLA, insurance update. This is never automatic.
- **Bolt/Uber fleet** — vehicle transfer after full payment requires physical handover, documentation, and re-registration

Automatic ownership transfer would be dangerous — there may be outstanding obligations (vehicle condition, contract terms, documentation) beyond just the payment amount.

Implementation Status: Done

- "Completed" status renamed to "Financially Complete" to avoid confusion
 - Ownership transfer is a separate manual/legal process (blocked pending Chairman meeting for the specific workflow)
 - `isFinished()` helper distinguishes between "payment done" and "agreement closed"
-

7. Vehicle Swap Workflow

Decision

Admin can swap a fleet driver's vehicle. This terminates the old agreement and creates a new one with the replacement vehicle. The admin chooses whether to carry over the amount already remitted.

Industry Standard

Vehicle swaps happen in fleet operations for:

- Vehicle maintenance or breakdown
- Vehicle upgrade or downgrade
- Driver performance (graduating to a better vehicle)

The terminate-and-recreate approach is standard:

- **Bolt fleet partners** — when a vehicle is swapped, a new arrangement is created. The fleet partner decides whether to credit previous payments.
- **Vehicle leasing companies** — lease termination + new lease agreement, with or without transfer of paid equity
- **Commercial fleet operators** — new assignment, new contract

The alternative (modifying the existing agreement to point to a new vehicle) would break the audit trail and make financial reconciliation difficult. A clean termination + new agreement preserves full history.

Implementation Status: Done

- Admin swap endpoint: terminates old agreement, unassigns old vehicle, assigns new, creates new agreement
 - Optional carry-over of total remitted amount (admin decision)
 - Optional new daily target and total cost (for different vehicle model)
 - Full audit log recording both old and new agreement IDs
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8. Termination with Settlement

Decision

When an agreement is terminated early, the admin records:

- Vehicle return status (returned, pending return, not returned, damaged)
- Outstanding amount owed
- Settlement amount agreed
- Settlement notes

Industry Standard

Structured settlement recording is standard in vehicle finance and fleet operations:

- **Hire-purchase termination** — the Hire Purchase Act requires accounting for: amount paid, amount outstanding, value of goods returned, and any settlement agreed
- **Vehicle leasing** — early termination agreements always document the financial reconciliation
- **Fleet operators** — vehicle return condition affects settlement (damage deductions, outstanding fees)

Recording this data digitally rather than on paper/spreadsheet is a maturity marker that financial auditors and regulators expect from technology platforms.

Implementation Status: Done

- Termination with settlement endpoint capturing all financial details
 - Vehicle return status tracking (returned, pending, not returned, damaged)
 - Outstanding and settlement amount recording
 - Free-text settlement notes for context
 - Audit log for every termination with settlement
-

9. Daily Settlement at 4:00 AM WAT

Decision

The system settles each day's remittance records at 4:00 AM West Africa Time. This time is configurable via environment variable.

Industry Standard

- **Bolt** — daily settlement window in the early morning hours
- **Banking/fintech** — end-of-day reconciliation typically runs between 2-5 AM
- **Ride-hailing platforms** — daily earnings calculated in the early morning for the previous day

4:00 AM WAT is after the latest reasonable ride in most Nigerian cities and before the earliest morning commuters, making it an appropriate cut-off.

Implementation Status: Done

- Configurable reset time via `config/fleet.php` and environment variable (`FLEET_DAILY_RESET_TIME`)
 - Two scheduled jobs at reset time: `DailyRemittanceSettlementJob` and `RemittanceShortfallAlertJob`
 - Timezone-aware scheduling (`Africa/Lagos`)
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10. Audit Trail for All Financial Operations

Decision

Every financial operation — agreement creation, updates, terminations, settlements, vehicle swaps, excused days, shortfall escalations — creates an audit log entry.

Industry Standard

Comprehensive audit logging is required by Nigerian financial regulations:

- **CBN Consumer Protection Framework** — requires records of all customer financial transactions
- **FCCPA** — requires businesses to maintain records of consumer transactions
- **ISO 27001 / SOC 2** — security and compliance frameworks require audit trails for financial operations

For a platform handling daily financial transactions between the company and drivers, a complete audit trail is not optional — it is a regulatory and operational necessity.

Implementation Status: Done

- Audit log entries for: agreement creation, updates, pauses, resumes, terminations, settlements, vehicle swaps, excused days, shortfall flag changes
- Each entry records: who performed the action, when, and the relevant details

11. What Is Deferred and Why

Feature	Reason for Deferral	Industry Precedent
Ownership transfer workflow	Awaiting Chairman meeting (Q8)	Standard: separate legal/admin process; not automated
Cash overpayment → wallet credit	Built — internal ledger credit (no external provider dependency)	Standard: closed-loop wallet credit via internal ledger
Fleet maintenance module	Phase 2 — separate product domain	Standard: fleet management platforms (Fleetio, Samsara) treat this as a separate module
Independent driver onboarding	Client said later phase	Standard: fleet and independent drivers are different compliance tracks
EV integration	No EV fleet yet	Standard: EV telemetry is a separate integration layer (API partnerships with vehicle OEMs)

What We Are NOT Building (Per Chairman's Directive)

The client originally asked to "code everything but leave it inactive." Per the Chairman's instruction to follow standard practice:

- **We do not build inactive features.** Industry practice is to design for extensibility (our database schema and service interfaces support future features) but not build, test, and maintain code that has no user. Inactive code creates maintenance burden, misleading test coverage, and technical debt.
- **We build when needed.** Each feature listed above can be built when its prerequisites are met and the business is ready to use it. The architecture supports this without rework.

Part II — Decision Analysis & Risk Flags

12. Decision Summary (Q1–Q10)

Question	Decision	Status
Q1: Payment model	Fixed daily remittance target (¥40,000 default, configurable per agreement)	Needs confirmation on per-vehicle-type variance
Q2: Shortfall handling	Tiered 3-7-14 day escalation: warning → review → escalated	Decided and implemented
Q3: Vehicle swapping	Admin-initiated, terminate + recreate, optional carry-over	Decided and implemented
Q4: Early termination	Vehicle returned, settlement recorded, HR/admin determines details	Decided and implemented
Q5: Deduction timing	Remittance-first per ride — not end-of-day batch	Decided and implemented
Q6: Commission rate	Zero commission for fleet drivers; all revenue from remittance	Decided and implemented
Q7: Price transparency	Reversed — drivers see full financial terms; UI emphasises percentage	Decided and implemented
Q8: Ownership transfer	Deferred — meeting with Chairman required	Blocked
Q9: Multi-vehicle	One vehicle per driver (Phase 1); external drivers later	Decided
Q10: Insurance & maintenance	E-tiGo covers everything; costs bundled into daily remittance	Decided (module deferred to Phase 2)

13. Decision-by-Decision Analysis

Q1: Daily Payment Amount — Fixed or Variable?

Status: Not directly answered, but Q5 implies a **fixed daily remittance target of ¥40,000**.

Action needed: Confirm this is the correct amount and whether it varies by vehicle type/model.

Implementation note: The ¥40,000 is configurable per agreement — different vehicle models can have different daily targets. Default is set via `config/fleet.php`.

Q2: Shortfall Handling

Decision: Tiered 3-7-14 day escalation with excused-day support.

Implementation: Complete — automatic streak tracking, configurable thresholds, excused-day admin endpoint, `RemittanceShortfallAlertJob` runs daily.

Q3: Vehicle Swapping

Decision: Admin-initiated only. Payment history optionally transfers via carry-over.

Implementation: Complete — `POST /admin/fleet-agreements/{id}/swap` with carry-over option. Terminates old, creates new, full audit log.

Q4: Early Termination

Decision: Vehicle returned, settlement recorded, HR/admin determines financial details.

Implementation: Complete — `POST /admin/fleet-agreements/{id}/terminate-settle` with vehicle return status, outstanding amount, settlement amount, and notes.

What the system cannot do: Determining send-off amounts or fault in accidents — these are human/HR/legal decisions. The system provides the data and records the outcome.

Q5: Deduction Timing — Remittance-First Per Ride

Decision: After each ride, the fare counts toward the daily remittance target. Once met, subsequent earnings go to the driver.

Implementation: Complete — wired into `ProcessPaymentJob`, automatic per-ride recording.

Cash ride handling: Cash rides count toward remittance target. The cash overpayment → rider wallet credit feature is now **implemented**. When a driver confirms cash collection with `amount_collected` greater than the fare, the overpayment is credited to the rider's wallet via an internal double-entry ledger journal (debit `DriverEarningsAvailable`, credit `PassengerWallet`). No external payment provider API call is required — the feature operates entirely within the existing ledger system. Configurable limits: max overpayment ₦2,000, respects existing wallet max balance ₦500,000.

Q6: Commission Rate for Fleet Drivers

Decision: Zero commission. All platform revenue from the daily remittance.

Implementation: Complete — enforced in `CommissionService@getRate()`.

Unit economics note: If a driver works 26 days/month at ₦40,000/day = ₦1,040,000/month revenue per vehicle. The client should validate this covers: vehicle cost amortisation, insurance, maintenance, permits, E-tiGo's margin, depreciation, and accident reserve.

Q7: Vehicle Price Transparency

Original decision: Hide total vehicle price from drivers.

Reversed to: Show full financial terms, emphasise percentage in UI.

Implementation: Complete — `total_vehicle_cost` and `remaining_amount` visible to all users. See [Section 5](#) for industry standard justification.

Q8: Ownership Transfer

Decision: Deferred. Meeting required with Chairman (Engr. Sylvester Unokesan).

Impact: Ownership transfer workflow cannot be built until the specific process is defined. The "Financially Complete" status is in place — this is the handoff point.

Q9: Multi-Vehicle / External Drivers

Decision: One vehicle per driver. Fleet-only for Phase 1. External (driver-owned) vehicles later with requirements: vehicle inspections, Certificate of Directorate of Road Traffic Services, E-tiGo driver ID, hackney permit, driver certification, e-hailing operator registration license.

Implementation: One vehicle per driver is already enforced. Certification tracking will be built when external drivers are introduced.

Q10: Insurance & Maintenance

Decision: E-tiGo insures all fleet vehicles (cost built into daily remittance). Routine maintenance included. Accident excess/deductible system for at-fault accidents.

Implementation: The daily remittance amount already includes insurance and maintenance — no separate tracking needed unless itemised breakdowns are requested. Fleet maintenance module deferred to Phase 2.

14. Items That Are NOT Codebase-Enforceable

These decisions require human processes, legal agreements, or physical-world actions. The platform **supports** them with data, workflows, and admin tools, but cannot **enforce** them.

Item	Why It's Not Code-Enforceable	What the System CAN Do
Send-off money determination (Q4)	HR/admin judgment call	Provide termination workflow with financial summary; admin enters the send-off amount
Accident fault determination (Q4, Q10)	Legal/insurance assessment	Record incident, link to driver/vehicle, track outcome entered by admin
Vehicle inspection and physical certification (Q9)	Physical process	Track certification documents (uploaded + admin-approved), expiry dates, block going online if expired
Insurance claims processing (Q10)	External insurer process	Record claims, link to incidents, track status
Contract terms and hire-purchase agreement (Q4)	Legal document	Store agreement reference and key dates; cannot enforce contract clauses
Driver negligence assessment (Q10)	Human judgment + legal	Provide incident data; admin records the outcome
Determining if a swap is warranted (Q3)	Admin board decision	Provide driver history, vehicle status, agreement details; admin initiates the swap
Vehicle DVLA re-registration (Q8, Q9)	Government process	Track registration status and expiry

Hackney permits, e-hailing licenses (Q9)	Regulatory bodies	Track document uploads and approval status
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15. Risk Flags

FLAG 1: Hiding Vehicle Price (Q7) — RESOLVED

Original risk: Hiding the total vehicle price would violate the Hire Purchase Act and FCCPA disclosure requirements.

Resolution: Q7 reversed. Drivers now see full financial terms. **This risk is eliminated.**

FLAG 2: Cash Change → Rider Wallet Credit (Q5) — RESOLVED & IMPLEMENTED

Original risk: Creating stored value in the rider's wallet may require a licence under CBN Guidelines on Electronic Money (2015).

Resolution: The feature uses the existing internal double-entry ledger — no external stored-value creation is involved. When a driver reports collecting more cash than the fare, the overpayment is recorded as a balanced journal entry (debit DriverEarningsAvailable, credit PassengerWallet). This is closed-loop platform credit, not e-money issuance. The driver's balance adjusts at settlement (same pattern as commission on cash rides). **This risk is eliminated.** Feature is live with configurable guardrails (max overpayment ₦2,000, wallet balance cap ₦500,000).

FLAG 3: Zero Commission Model (Q6) — AWARENESS ONLY

Not a risk — this is a valid business model. But the ₦40,000 daily target must cover all costs (vehicle, insurance, maintenance, margin, depreciation, accident reserve). This is a business modelling exercise, not a code decision.

FLAG 4: Remittance-First Model — Driver Experience

Not a risk — the model is industry standard. However, on slow days a driver may earn zero take-home pay. The tiered escalation with excused days mitigates this. Awareness for driver retention strategy.

16. Earnings Model

Good Day Scenario

Day Summary:	
Total rides:	14
Total fares:	₦62,000
Remittance (E-tiGo):	₦40,000
Driver take-home:	₦22,000
Platform commission:	₦0 (zero for fleet)
Shortfall:	₦0

Slow Day Scenario

Day Summary:

Total rides:	6
Total fares:	₦28,000
Remittance (E-tiGo):	₦28,000 (partial – target was ₦40,000)
Driver take-home:	₦0
Platform commission:	₦0
Shortfall recorded:	₦12,000

Part III — Technical Implementation

17. Data Model

fleet_agreements — hire-to-own agreement per driver-vehicle pair

Column	Type	Description
id	UUID PK	
driver_id	FK → drivers	Driver on this agreement
vehicle_id	FK → vehicles	Fleet vehicle assigned
daily_remittance_target	decimal(12,2)	E.g., 40000.00 (configurable per agreement)
total_vehicle_cost	decimal(14,2)	Full purchase price (visible to driver per Q7 reversal)
total_remitted	decimal(14,2)	Running sum of all remittance collected
agreement_start_date	date	When the agreement started
status	string	active, paused, completed, terminated
shortfall_streak_days	integer	Consecutive days with shortfall
shortfall_flag	string nullable	warning, review, or escalated
shortfall_flagged_at	timestamp nullable	When the flag was last set
vehicle_return_status	string nullable	returned, pending_return, not_returned, damaged
outstanding_amount	decimal(14,2) nullable	Outstanding amount on termination
settlement_amount	decimal(14,2) nullable	Settlement amount agreed
settlement_notes	text nullable	Free-text settlement details
settled_at	timestamp nullable	When settlement was recorded
terminated_reason	text nullable	If terminated: why

terminated_at	timestamp nullable	
completed_at	timestamp nullable	When fully paid off
paused_at	timestamp nullable	When paused
created_by_admin_id	FK → users	Admin who created the agreement
timestamps		

Indexes: (driver_id, status) , (vehicle_id, status)

progress_percentage is computed at runtime: $(\text{total_remitted} / \text{total_vehicle_cost}) * 100$

daily_remittances — daily record per driver

Column	Type	Description
id	UUID PK	
agreement_id	FK → fleet_agreements	
driver_id	FK → drivers	
date	date	The remittance day
target_amount	decimal(12,2)	Daily target (snapshot from agreement)
remitted_amount	decimal(12,2)	Total collected toward target this day
shortfall_amount	decimal(12,2)	target - remitted (0 if target met)
target_met_at	timestamp nullable	When the target was met (null if not met)
ride_count	integer	Number of rides that day
total_fares	decimal(12,2)	Total fare value of all rides
driver_earnings	decimal(12,2)	What the driver kept (fares after target met)
settled	boolean	Whether end-of-day settlement is complete
excused_reason	text nullable	Why this day was excused (e.g., "vehicle_downtime: in maintenance")
timestamps		

Indexes: (agreement_id, date) UNIQUE, (driver_id, date) , (settled, date)

18. Service Layer

FleetRemittanceService

Method	Description
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<code>createAgreement()</code>	Create new agreement with validation (no duplicate active, fleet vehicle required)
<code>updateAgreement()</code>	Update daily target (active only), audit log with old/new values
<code>terminateAgreement()</code>	Basic termination with reason
<code>terminateWithSettlement()</code>	Termination with vehicle return status and financial reconciliation
<code>pauseAgreement()</code> / <code>resumeAgreement()</code>	Toggle active/paused with audit log
<code>recordRideRemittance()</code>	Per-ride allocation with row-level locking; auto-completes when total reached
<code>settleDay()</code>	End-of-day settlement; skips excused days; updates shortfall streak
<code>excuseDay()</code>	Mark remittance day as excused; decrements streak by 1; audit log
<code>swapVehicle()</code>	Terminate old agreement, unassign old vehicle, assign new, create new agreement
<code>getOrCreateTodayRemittance()</code>	Gets or creates today's daily remittance record

CommissionService

- `getRate()` returns 0.0 for drivers with active fleet agreements (early return before override/global/default logic)

ProcessPaymentJob

- After payment processing, calls `FleetRemittanceService@recordRideRemittance()` for fleet drivers

19. Background Jobs

Job	Schedule	Description
DailyRemittanceSettlementJob	Daily at 4:00 AM WAT	Ensures every active agreement has a remittance row for the date, then settles all unsettled records
RemittanceShortfallAlertJob	Daily at 4:00 AM WAT	Checks active agreements against thresholds, assigns shortfall flags, creates audit logs

20. Configuration

File: `config/fleet.php`

Key	Default	Env Variable	Description
<code>daily_reset_time</code>	04:00	<code>FLEET_DAILY_RESET_TIME</code>	When daily settlement runs

shortfall_warning_days	3	FLEET_SHORTFALL_WARNING_DAYS	Days before warning flag
shortfall_review_days	7	FLEET_SHORTFALL_REVIEW_DAYS	Days before review flag
shortfall_escalation_days	14	FLEET_SHORTFALL_ESCALATION_DAYS	Days before escalated flag
default_daily_target	40000	FLEET_DEFAULT_DAILY_TARGET	Default daily remittance target

Part IV — Developer Handoff (API & UI/UX)

For: Designer, Frontend, and Mobile Teams

Base URL: {APP_URL}/api/v1

21. API Endpoints — Driver (3)

These are what the **mobile app** consumes.

1. Today's Remittance Progress

```
GET /driver/remittance/today
Authorization: Bearer {driver_token}
```

Returns the real-time remittance progress for today. Primary data source for the driver's daily dashboard.

Response fields:

Field	Type	Description
target_amount	string (decimal)	Daily target, e.g. "40000.00"
remitted_amount	string (decimal)	Amount collected toward target so far
shortfall_amount	string (decimal)	How much is still needed (target - remitted, 0 if met)
target_met	boolean	true once daily target is fully met
target_met_at	datetime / null	Timestamp when target was met
ride_count	integer	Number of rides completed today
total_fares	string (decimal)	Sum of all ride fares today
driver_earnings	string (decimal)	What the driver has earned AFTER target was met
excused_reason	string / null	If day is excused, shows the reason

Returns {"remittance": null} if the driver has no active fleet agreement.

2. Remittance History

```
GET /driver/remittance/history
Authorization: Bearer {driver_token}
```

Paginated list of daily remittance records, most recent first. Same fields as today's endpoint, plus `settled` (boolean) and `date` .

Standard pagination via `meta block ( current_page , last_page , per_page , total )`.

3. Fleet Agreement

```
GET /driver/fleet-agreement
Authorization: Bearer {driver_token}
```

Returns the driver's active fleet agreement summary.

Response fields:

Field	Type	Description
daily_remittance_target	string (decimal)	The daily amount they need to remit
total_vehicle_cost	string (decimal)	Full vehicle purchase price
total_remitted	string (decimal)	Total amount remitted across all days
remaining_amount	string (decimal)	How much is left to pay
progress_percentage	float	0-100, progress toward vehicle ownership
agreement_start_date	date string	When the agreement started
status	string	active, paused, completed, terminated
shortfall_streak_days	integer	Consecutive days where target was not met
vehicle	object	Vehicle details (make, model, plate, etc.)

Design note: Drivers see `total_vehicle_cost` and `remaining_amount` alongside `progress_percentage` . The dashboard should emphasise percentage progress but the driver must have access to the full financial terms per hire-purchase disclosure requirements.

Returns `{ "agreement": null }` if no active fleet agreement.

22. API Endpoints — Admin (11)

Middleware: `auth:sanctum , user.type:admin , admin.role:operations,finance`

Method	Endpoint	Description
GET	<code>/admin/fleet-agreements</code>	List all agreements (filter by <code>status</code> , <code>driver_id</code>)
POST	<code>/admin/fleet-agreements</code>	Create new agreement

GET	/admin/fleet-agreements/{id}	Agreement detail + last 30 remittance records
PUT	/admin/fleet-agreements/{id}	Update daily target (active only)
POST	/admin/fleet-agreements/{id}/terminate	Terminate with reason
POST	/admin/fleet-agreements/{id}/terminate-settle	Terminate with full settlement details
POST	/admin/fleet-agreements/{id}/pause	Pause (e.g., vehicle maintenance)
POST	/admin/fleet-agreements/{id}/resume	Resume paused agreement
POST	/admin/fleet-agreements/{id}/swap	Swap vehicle (terminates old, creates new agreement)
POST	/admin/fleet-agreements/{id}/remittances/{remittance_id}/excuse	Mark a day as excused

Admin responses include additional fields: `shortfall_flag` , `shortfall_flagged_at` , `vehicle_return_status` , `outstanding_amount` , `settlement_amount` , `settlement_notes` .

Creating an Agreement (Admin Flow)

1. Mark a vehicle as fleet: `PATCH /admin/drivers/{driver_id}/vehicle/fleet`
2. Create the agreement: `POST /admin/fleet-agreements` with:
 - `driver_id` (UUID)
 - `vehicle_id` (UUID, must be `is_fleet: true`)
 - `daily_remittance_target` (number, min 1000)
 - `total_vehicle_cost` (number, min 100000)
 - `agreement_start_date` (date, today or future)

A driver can only have **one active agreement** at a time.

Swap Fleet Vehicle

`POST /admin/fleet-agreements/{id}/swap`

Field	Type	Required	Description
<code>new_vehicle_id</code>	UUID	Yes	Replacement fleet vehicle (must be unassigned)
<code>reason</code>	string	Yes	Reason for the swap (max 1000)
<code>carry_over_remitted</code>	boolean	Yes	Whether to carry over total_remitted
<code>daily_remittance_target</code>	number	No	New daily target (defaults to previous)
<code>total_vehicle_cost</code>	number	No	New total cost (defaults to previous)

Terminate with Settlement

```
POST /admin/fleet-agreements/{id}/terminate-settle
```

Field	Type	Required	Description
reason	string	Yes	Reason for termination (max 1000)
vehicle_return_status	string	No	returned, pending_return, not_returned, damaged
outstanding_amount	number	No	Outstanding amount owed
settlement_amount	number	No	Settlement amount agreed
settlement_notes	string	No	Free-text notes (max 2000)

Excuse Remittance Day

```
POST /admin/fleet-agreements/{id}/remittances/{remittance_id}/excuse
```

Field	Type	Required	Description
reason	string	Yes	approved_leave, vehicle_downtime, low_demand, payment_failure, other
notes	string	No	Additional context (max 1000)

23. UI/UX Guidance — Driver App

1. Daily Remittance Dashboard (Primary Screen for Fleet Drivers)

This should be prominently visible when the driver is online. Think of it like a fuel gauge.

States to handle:

State	Visual Suggestion
Target not yet met	Progress bar/circle showing remitted_amount / target_amount. Show "¥28,500 of ¥40,000"
Target met	Celebration moment. Switch to "You're earning! ¥X earned so far"
No agreement	Hide the remittance widget entirely — this is a regular (non-fleet) driver

Data to display:

- Progress toward daily target (bar, circle, or percentage)
- Number of rides completed today
- Amount still needed: shortfall_amount
- Driver earnings (only meaningful after target met)
- Time target was met (target_met_at) — e.g., "Target met at 2:22 PM"

2. Remittance History

Calendar or list view showing daily records. Each day shows:

- Date
- Target met (yes/no — use checkmark/cross)
- Amount remitted vs target
- Number of rides
- Driver earnings for that day
- Shortfall (if any, highlighted)
- Excused days marked distinctly (grey/neutral, not red)

Consider colour-coding: green for target met, red/amber for shortfall days, grey for excused days.

3. Fleet Agreement Summary

A card or screen showing:

- Vehicle info (make, model, plate)
- Daily remittance target
- Progress toward ownership (percentage — primary display)
- Total vehicle cost, total remitted, remaining amount (accessible via "View details" or expandable section)
- Agreement status
- Shortfall streak (if > 0, show as a warning)
- Agreement start date

Design note: Lead with the percentage progress bar. The full financial details should be accessible but not the default view. This satisfies both the client's UX preference (percentage emphasis) and legal disclosure requirements.

24. UI/UX Guidance — Admin Dashboard

1. Fleet Agreements List

Table/list view with columns:

- Driver name
- Vehicle (make + plate)
- Daily target
- Total remitted / Total cost (admin sees both)
- Progress %
- Status (with colour badges: green=active, yellow=paused, grey=completed, red=terminated)
- Shortfall streak (flag if > 0)
- Shortfall flag (warning/review/escalated — colour-coded)

Filters: status dropdown, driver search

2. Fleet Agreement Detail

- All financial data (admin sees everything including settlement details)
- Recent remittance records (last 30 days shown by default)
- Action buttons: Pause, Resume, Terminate, Swap, Terminate with Settlement (context-dependent — only show valid actions for current status)
- Excuse day action on individual remittance records
- Audit trail (agreement history)

3. Create Agreement Form

Fields:

- Driver selector (searchable dropdown)
- Vehicle selector (filtered to fleet vehicles only)
- Daily remittance target (number input, default 40000)
- Total vehicle cost (number input)
- Agreement start date (date picker, today or future)

Validation messages to handle:

- "Driver already has an active fleet agreement"
- "Vehicle must be marked as fleet before creating an agreement"

25. Enum Values

Fleet Agreement Status

Value	Label	Description
active	Active	Agreement is in force, remittance is being tracked
paused	Paused	Temporarily suspended (e.g., vehicle maintenance, driver leave)
completed	Financially Complete	Total vehicle cost fully remitted — ownership transfer is a separate manual process
terminated	Terminated	Agreement ended early by admin (with reason)

Shortfall Flag

Value	Trigger	Description
warning	3+ consecutive shortfall days	Initial alert to admin
review	7+ consecutive shortfall days	Admin review required
escalated	14+ consecutive shortfall days	Escalation — action required

Vehicle Return Status

Value	Description
returned	Vehicle physically returned to E-tiGo
pending_return	Return agreed but not yet completed
not_returned	Vehicle not returned
damaged	Vehicle returned in damaged condition

Excused Day Reasons

Value	Description
approved_leave	Driver on approved leave
vehicle_downtime	Vehicle under maintenance or repair
low_demand	Low ride demand (admin discretion)
payment_failure	Payment system issue
other	Other reason (requires notes)

26. Key Business Rules for UI

- 1. Fleet drivers earn #0 until daily target is met.** The UI should make this crystal clear — show it as "Working toward today's target" not "Earnings: #0".
- 2. Emphasise percentage progress but show full financial terms.** The UI should lead with percentage (e.g., "25% toward ownership") but the driver must be able to view total vehicle cost, remaining amount, and total remitted. This is a legal requirement for hire-purchase agreements.
- 3. One active agreement per driver.** The create agreement form should validate this and show a clear error if violated.
- 4. Fleet vehicles skip plate KYC verification.** When a vehicle is marked as fleet, the KYC flow changes — NIN + Driver's License is sufficient (no plate verification needed since E-tiGo owns the vehicle).
- 5. Zero commission for fleet drivers.** The earnings display should NOT show any commission deduction line. Fleet drivers see only: remittance vs. personal earnings.
- 6. Agreement auto-completes financially.** When `total_remitted >= total_vehicle_cost`, the agreement status changes to `completed` (financially complete). This does NOT mean ownership has transferred — that is a separate administrative and legal process. The driver should see a "Payment obligations fulfilled" state, not an ownership transfer celebration.

Part V — Status & Next Steps

27. Implementation Status

Phase 1 (Complete)

Feature	Status
Remittance agreement model (configurable target per agreement)	Done
Per-ride remittance accumulation and daily tracking	Done
Admin: create/edit/terminate/pause/resume agreements	Done
Shortfall tracking and recording	Done

Zero commission for fleet drivers	Done
Driver: today's remittance, history, agreement endpoints	Done
Daily settlement job	Done

Phase 1.5 (Complete — October 2026)

Feature	Status
Fleet remittance wired into ride completion flow	Done — automatic per-ride recording
Daily settlement cron at 4:00 AM WAT (configurable)	Done
3-7-14 day shortfall escalation with auto-flagging	Done
Excused-day support (leave, downtime, low demand)	Done
Vehicle swap endpoint with carry-over option	Done
Termination with settlement workflow	Done
Zero commission enforcement in CommissionService	Done
Full financial terms visible to drivers (Q7 reversal)	Done
Fleet configuration file (reset time, thresholds)	Done
Cash overpayment → rider wallet credit	Done — internal ledger credit, driver sends amount_collected on cash confirm

Not Yet Built

Feature	Status	Blocked By	Impact on UI
Ownership transfer workflow	Blocked	Q8 — Chairman meeting	Don't build a "Transfer ownership" button yet
Fleet maintenance module	Deferred	Phase 2	No vehicle servicing/maintenance screens yet
Independent driver onboarding	Deferred	Client said later phase	No own-vehicle driver compliance workflow yet
EV integration	Deferred	Architecture supports future extension	No EV-specific features yet

28. Testing & Postman

Test suite: 693 tests passing (684 passed, 9 pre-existing skips unrelated to fleet).

Postman Collection

Import `doc/postman_collection.json` — folders:

- **Admin — Fleet Agreement Management** (11 requests with example responses)
- **Driver — Fleet Remittance** (3 requests with example responses for all states)

Postman Variables Needed

Variable	Description	Example
<code>admin_token</code>	Admin auth token	From admin login
<code>driver_token</code>	Driver auth token	From driver OTP verify
<code>fleet_agreement_id</code>	UUID of a fleet agreement	From create or list response
<code>remittance_id</code>	UUID of a daily remittance	From agreement detail response
<code>driver_uuid</code>	UUID of a driver	From admin driver list
<code>vehicle_uuid</code>	UUID of a fleet vehicle	From driver vehicle endpoint

Quick Test Flow

1. Login as admin → get `admin_token`
2. Login as driver → get `driver_token`
3. Mark vehicle as fleet: PATCH `/admin/drivers/{id}/vehicle/fleet`
4. Create agreement: POST `/admin/fleet-agreements`
5. Check driver view: GET `/driver/fleet-agreement` (should show agreement with financial terms)
6. Check today: GET `/driver/remittance/today` (should be null until a ride happens)

Full API Documentation

See `doc/API_REFERENCE.md` — sections:

- **Admin — Fleet Agreement Management**
- **Driver — Fleet Remittance**
- **Enums** → **Fleet Agreement Status**

All endpoints include full request/response schemas with example payloads.

29. Outstanding Items & Next Steps

Client Action Required

1. **Confirm Q1:** Daily remittance amount (¥40,000 is currently configurable per agreement) and whether it varies by vehicle type
2. **Schedule Q8:** Meeting with Chairman on ownership transfer process

Resolved

- ~~Confirm payment provider:~~ Paystack selected (October 2026). PSSP licence covers wallet credit. OPay transfer support for driver payouts.

Resolved Engineering

- ~~Cash overpayment~~ → ~~wallet credit:~~ Implemented (11 October 2026). Uses internal double-entry ledger — no external provider dependency. Driver sends `amount_collected` with cash confirmation; overpayment auto-credited to rider's `PassengerWallet` .

Engineering Next

1. **Phase 2 planning:** Fleet maintenance module, independent driver onboarding

Legal Review

- Hire-purchase price disclosure: Q7 has been reversed — drivers now see full financial terms (compliant)
- Hire-purchase agreement template review recommended before launch

This document was last updated on 11 October 2026. Cash overpayment → wallet credit feature implemented. It consolidates the Fleet Industry Standards, Fleet Decisions Report, and Fleet Feature Handoff into a single reference. All referenced regulations (Hire Purchase Act, FCCPA, CBN guidelines) should be reviewed by E-tiGo's legal counsel before commercial launch.

Questions? Ping the backend team. The Postman collection has all endpoints ready for testing.